



**UNIVERSITY SCHOOL OF AUTOMATION AND ROBOTICS EAST
DELHI CAMPUS, SURAJMAL VIHAR, DELHI- 110032**

Minor Project Report

on

DeepFake Image & Video Detection Model

Submitted in partial fulfillment of
B.Tech (7th Semester)
In
Artificial Intelligence & Data Science

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**Under the supervision of
Mr. Anupam Lakhanpal
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CERTIFICATE BY SUPERVISOR

CERTIFICATE

This is to certify, that the project report submitted by Shreya Shree is an outcome of his work that was done during the Minor Project Development period. The work was original and done by the candidate. He has duly acknowledged all the sources from which the ideas and extracts have been taken.

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DECLARATION

I hereby declare that the Report entitled “**DeepFake Image & Video Detection Model**” is an authentic record of the work completed as part of the requirements of **Minor Project (ART455)** under the supervision of **Mr. Anupam Lakhanpal**.

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ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to my mentor, **Mr. Anupam Lakhanpal (Faculty Mentor)**, for providing me with the invaluable opportunity to undertake my minor project titled **“Deepfake Image and Video Detection Model .”** This project has been a highly enriching experience that allowed me to explore the rapidly growing domain of deepfake detection, machine learning, and web development through practical implementation and continuous guidance.

I extend my sincere thanks to **Mr. Anupam Lakhanpal (Faculty Mentor)** for his constant support, insightful feedback, and expert supervision throughout the duration of my project. His mentorship played a crucial role in helping me understand the technical, ethical, and real-world significance of deepfake detection, and enabled me to successfully develop a functional ML model along with an integrated website.

This project provided me with the opportunity to design and implement a machine learning model capable of detecting deepfake images and videos, along with a user-friendly web interface for testing and demonstrations. Working on the model training, preprocessing pipelines, evaluation metrics, and deployment strengthened my analytical thinking, problem-solving abilities, and practical understanding of AI-based security applications.

Finally, I would like to express my sincere appreciation to my university faculty members, friends, and family for their continuous motivation and support, which encouraged me to perform to the best of my potential throughout this project journey.

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ABSTRACT

The Minor Project provided an excellent opportunity to explore the intersection of artificial intelligence, computer vision, and digital media security. The project focused on understanding deepfake technology, feature extraction techniques, image manipulation patterns, and machine learning-based detection methods. Throughout the development process, I worked extensively with modern tools and frameworks such as MesoNet for classification, MTCNN for face detection, and MediaPipe for landmark analysis, gaining practical experience in building reliable deepfake detection systems.

As part of this project, I developed a model titled **“Deepfake Image and Video Detection System,”** designed to identify manipulated facial content in both images and videos. The objective of the project was to detect inconsistencies introduced during deepfake generation by analyzing facial features, spatial distortions, and frame-level artifacts. The model integrates MTCNN-based face extraction, MesoNet-based classification, and MediaPipe-driven landmark analysis to enhance detection accuracy. Additionally, a simple website was created using HTML and CSS to provide a user-friendly interface for uploading and analyzing media files.

The project workflow involved data collection, face preprocessing, training and evaluation of the deepfake detection model, followed by the design of an accessible frontend platform for demonstrations. The system outputs key indicators such as probability of manipulation and frame-wise detection scores, enabling effective analysis of suspicious media content.

This project significantly strengthened my understanding of machine learning, neural networks, image processing, and secure AI applications. It improved my practical skills in model development, computer vision workflows, and interface design, while highlighting the importance of technological solutions in combating misinformation and ensuring digital media authenticity.

CHAPTER 1: INTRODUCTION

The Minor Project was an intensive and application-oriented research and development experience designed to build practical knowledge in artificial intelligence, computer vision, and digital media forensics. The project provided both theoretical understanding and hands-on implementation, focusing on the detection of deepfake images and videos using modern machine learning techniques and face analysis frameworks. Throughout the project, I explored the complete workflow of deepfake detection — from data collection, preprocessing, model development, evaluation, to interface creation — enabling the application of technical concepts to address an important real-world cybersecurity challenge.

The learning and development process combined literature review, experimentation with ML models, and system design, resulting in a robust deepfake detection pipeline and a functional website for user interaction.

Key Learning Areas and Components Covered

1. Introduction to Deepfakes and Media Forensics:

- Understanding deepfake technology, its working mechanism, and its social, ethical, and security implications.
- Learning about common manipulation techniques in images and videos.
- Exploring the growing importance of AI-based detection for digital media authenticity.

2. Computer Vision & Face Detection Fundamentals:

- Gaining insights into core concepts of face detection, feature extraction, and landmark tracking.

- Understanding the role of convolutional neural networks (CNNs) in detecting manipulation artifacts.

(1)

- Studying spatial and temporal inconsistencies in videos.

3. Tools, Techniques, and Frameworks:

- Working with **MesoNet**, a lightweight CNN model effective for deepfake image detection.
- Using **MTCNN** (Multi-Task Cascaded Convolutional Network) for accurate face detection and cropping.
- Implementing **MediaPipe Face Mesh** for landmark-based artifact detection and analysis.
- Creating a simple front-end website using **HTML and CSS** for user interaction and model testing.

4. Data Collection and Preprocessing:

- Preparing datasets of real and fake images/videos for training and evaluation.
- Extracting faces using MTCNN to ensure consistent and high-quality inputs for the model.
- Applying preprocessing techniques such as resizing, normalization, and noise reduction.
- Organizing frame-wise datasets for video-level analysis.

5. Model Training, Evaluation, and Optimization:

- Training the MesoNet model on curated deepfake datasets to classify real vs. fake inputs
- Using MediaPipe landmarks to identify distortion patterns and enhance detection accuracy.

- Applying evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrices.

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- Optimizing model performance through hyperparameter tuning and improved preprocessing.

6. System Design and Visualization:

- Designing a clear and interpretable output structure showing manipulation probability and detection results.
- Implementing frame-wise detection to analyze videos effectively.
- Creating a clean and user-friendly website layout to upload media files and view results.

7. Real-World Application and Project Integration:

- Integrating all modules into a cohesive deepfake detection system.
- Developing the final project titled **“Deepfake Image and Video Detection Model and Website”** under the guidance of **Prof. Anupam Lakhanpal**.
- The project enhanced my understanding of computer vision, machine learning pipelines, and secure AI applications to combat misinformation and ensure digital media integrity.

CHAPTER 2: LITERATURE SURVEY

- **Deep Learning Architectures for Deepfake Detection :-**

Recent literature highlights that deepfake detection relies heavily on advanced deep learning techniques capable of identifying subtle visual and temporal inconsistencies. Deepfakes are primarily generated using **autoencoders, GANs, and face-swapping algorithms**, which makes detection challenging as the synthetic outputs keep improving.

Modern research emphasizes the use of specialized CNN architectures such as **MesoNet**, spatial-temporal frameworks such as **MTCNN**, and feature-processing libraries such as **MediaPipe** for robust deepfake identification.

MesoNet

A lightweight convolutional neural network designed specifically to detect facial manipulation by capturing mid-level features such as blending artifacts, texture inconsistencies, and boundary irregularities.

MTCNN (Multi-task Cascaded Convolutional Networks)

Primarily cited as one of the most efficient face detection and alignment models. It enables:

- Accurate face cropping
- Landmark detection
- Boundary alignment

These serve as crucial preprocessing steps before classification.

MediaPipe

A high-performance face processing framework used for:

- Real-time face mesh extraction
- Facial landmark tracking
- Temporal consistency analysis

MediaPipe helps detect unnatural motion patterns and micro-expressions found in deepfake videos.

Collectively, these models form a robust pipeline for detecting image and video-based deepfakes.

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- **Feature Engineering and Detection KPIs for Deepfake Identification :-**

Key Performance Indicators (KPIs) and detection metrics from the literature help quantify manipulation patterns in synthetic media. These KPIs focus on identifying inconsistencies in facial regions, temporal patterns, and visual artifacts.

Common KPIs include:

- **Artifact Detection:** Analysis of blending errors, color mismatch, and unnatural textures resulting from poor face-swapping or GAN synthesis.
- **Facial Landmark Deviation:** MediaPipe and MTCNN extract 468+ facial points to measure:
 - 🎬 **Misaligned boundaries**
 - 🎬 **Abnormal warping**
 - 🎬 **Inconsistent blinking or lip-sync patterns**
- **Temporal Consistency Score:** Used for videos to evaluate frame-to-frame coherence:
 - 🎬 **Sudden pixel shifts**
 - 🎬 **Unnatural head movements**
 - 🎬 **Missing physiological signals (eye movement, micro-expressions)**
- **CNN Classification Output:** MesoNet produces a probability score indicating how likely the input is fake vs. real.
- **Performance Metrics:** Evaluated through accuracy , Precision , Recall , F1-Score , AUC-ROC , Confusion Matrix.

These KPIs collectively help determine whether an image or video exhibits deepfake characteristics.

- **Deepfake Dataset Modeling and Preprocessing :-**

Literature shows that successful deepfake detection depends significantly on dataset design,

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modeling, and preprocessing. The typical dataset structure follows a **feature-extraction pipeline**:

- **Frame-Level Structuring:** Videos are split into individual frames (1–5 FPS) to create consistent training samples.
- **Face Detection & Alignment:** Using MTCNN, each frame undergoes face detection , cropping , alignment using eye and mouth landmarks. This ensures uniformity for model training.
- **Feature Extraction Using MediaPipe:** MediaPipe extracts Facial mesh , Temporal Feature Vectors , Motion dynamics. These vectors help models detect deepfake inconsistencies more effectively.
- **Model Training Setup:** A typical deepfake detection dataset contains **real images/videos** , **manipulated images/videos** which generated using **GAN-bases models** , **Autoencoders** , **Face-swapping techniques**.

This structured dataset enables reliable model training and generalization.

- **Deepfake Image and Video Forecasting & Classification :-**

While deepfake detection is not time-series based like wind energy forecasting, literature emphasizes *classification forecasting*, i.e., prediction of manipulation probability.

The classification model (MesoNet or CNN-based) predicts:

- **Probability of being fake (%)**
- **Feature inconsistency levels**
- **Confidence interval of detection**

These forecast-style outputs help users understand how strongly a model believes the media is manipulated.

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- **Website Implementation & Analytical Framework :-**

Recent studies highlight the importance of deploying deepfake detection systems through interactive web applications to ensure accessibility and real-time processing.

Key Web Framework Features include:

- **Upload & Detection Module:** User can upload Images and Videos. The backend pipeline extracts frames, applies models, and returns detection scores.
- **Real-time Visualization:** The website displays real or fake images/videos . It displays frame-by-frame analysis for videos.
- **Frontend:** Using HTML/CSS/JS ,it is used for UI design , progress indicators , graphical representation of outputs.
- **Backend:** Using Python , it handles frame extraction , preprocessing with MTCNN + MediaPipe , classification with MesoNet , response generation.

This turns the website into an informative detection portal rather than just a classifier.

CHAPTER 3: PROBLEM STATEMENT

In the digital era, the rapid advancement of artificial intelligence and generative models has led to the rise of **deepfakes**—highly realistic but artificially manipulated images and videos. While these technologies have legitimate applications in entertainment, education, and creative industries, they pose significant threats when misused. Deepfakes can spread **misinformation, identity fraud, defamation, political manipulation, and cybercrimes**, making automated deepfake detection an urgent global necessity.

The core challenge lies in the fact that modern deepfake generation techniques—powered by GANs, autoencoders, and advanced face-swapping algorithms—produce manipulations that are increasingly difficult for the human eye or traditional detection systems to identify.

Another major difficulty is the **lack of unified systems** capable of detecting deepfakes in both images and videos. Video deepfakes require frame-by-frame analysis, temporal tracking, and detection of micro-expression inconsistencies, whereas image deepfakes demand high spatial resolution analysis. Existing solutions often struggle with:

- Identifying subtle artifacts hidden in high-quality synthetic media.
- Processing real-time detection efficiently
- Handling diverse lighting conditions, poses, and facial angles
- Providing an accessible platform for non-technical users

Without an integrated and automated detection framework, individuals, organizations, and law enforcement agencies face significant challenges in verifying the authenticity of digital content and combating deepfake-driven cyber threats.

To address these challenges, this project aims to design and develop a **Deepfake Image and Video Detection Model** using **MesoNet, MTCNN, and MediaPipe**, along with a user-friendly **website interface built using HTML and CSS**.

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CHAPTER 4: DESCRIPTION OF VARIOUS TRAINING MODULES

The development of the **Deepfake Image and Video Detection System**, incorporating models such as **MesoNet, MTCNN, and MediaPipe**, along with a user-friendly website built using **HTML and CSS**, was structured into multiple training modules. These modules were designed to provide both theoretical knowledge and practical implementation skills in deep learning, computer vision, and web development. Each module gradually introduced fundamental concepts and advanced methodologies required to analyze, detect, and classify deepfake media, leading to the creation of a complete detection system and frontend platform.

Module 1: Introduction to Deepfakes and Digital Media Manipulation

The first module provided an overview of deepfake technology, its evolution, and its societal impact. Learners explored how deepfake images and videos are created using GANs, autoencoders, and face-swapping algorithms. The module emphasized the challenges deepfakes pose in misinformation, cybersecurity, identity theft, and privacy violation. Additionally, this module introduced the fundamental concepts of computer vision and deep learning required for image/video analysis.

Key areas covered:

- Understanding deepfake generation pipelines
- Types of deepfakes (face swap, lip-sync, identity swap)

- Threat landscape and real-world risks
- Overview of detection techniques and research trends

Module 2: Tools, Libraries, and Technologies for Deepfake Detection

This module introduced the technical tools required for developing the detection system. Students gained hands-on experience with computer vision libraries and deep learning frameworks essential for model training and evaluation.

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Key tools and technologies included:

- Python and OpenCV for preprocessing and face detection
- TensorFlow/Keras for deep learning model implementation
- MTCNN for face extraction and alignment
- MediaPipe for landmark detection and face mesh analysis
- Jupyter Notebook/Colab for model experimentation

The module also covered essential concepts such as GPU acceleration, dataset loading, and environment setup.

Module 3: Dataset Preparation, Preprocessing, and Frame Extraction

A significant part of the training focused on preparing datasets for deepfake detection. Students learned how to extract frames from videos, identify facial regions, and prepare labeled datasets for training and testing.

Topics covered in this module:

- Frame extraction from deepfake and real video datasets
- Face detection and cropping using MTCNN
- Landmark extraction and feature enhancement using MediaPipe
- Handling imbalanced datasets
- File structuring, resizing, and normalization

The module ensured that datasets were clean, consistent, and ready for deep learning models.

Module 4: Exploratory Data Analysis (EDA) and Feature Understanding

This module emphasized understanding both visual and statistical patterns in real vs. fake media. Through EDA, learners analyzed how deepfake artifacts manifest and how these patterns can be used for classification.

Key topics included:

- Visual artifact inspection (blurring, texture mismatches, boundary distortions)
- Temporal inconsistencies in video sequences
- Statistical differences between manipulated and genuine frames

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- Identifying features essential for deepfake detection

The goal was to build a strong foundation for selecting the right detection model and feature set.

Module 5: Deepfake Detection Models – MesoNet, CNNs, and Temporal Analysis

In this module, students implemented the core deepfake detection models.

The focus was on using lightweight and efficient architectures suitable for image and video-based detection.

Concepts covered:

- Implementing MesoNet for image-based deepfake detection
- Using CNNs for feature extraction and classification
- Applying temporal consistency checks for video deepfakes
- Training, validation, and hyperparameter tuning
- Evaluating model metrics (accuracy, precision, recall, F1-score)

The module demonstrated how spatial and temporal patterns enable reliable deepfake classification.

Module 6: Building and Integrating the Detection Pipeline

This module combined the preprocessing, modeling, and inferencing components into a functional detection pipeline.

Training included:

- Creating a modular detection system for images and videos

- Performing frame-by-frame analysis using MTCNN and MediaPipe
- Integrating deep learning models for inference
- Generating confidence scores and producing results

This module ensured the model could operate independently and efficiently, forming the backend logic of the system

Module 7: Website Development Using HTML and CSS

The final module focused on building the frontend interface to make the system accessible to users.

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Using HTML and CSS, students created a simple, responsive website where users could upload images or videos for analysis.

Key areas:

- Designing a clean and user-friendly interface
- Upload input fields, result display sections, and styling
- Ensuring compatibility with the backend detection model
- Basic UI/UX principles for security-focused tools

This module emphasized creating a functional web interface that bridges users with the deepfake detection system.

Through these structured and progressive training modules, students developed comprehensive expertise in **deep learning, computer vision, dataset engineering, model deployment, and web development**. The step-by-step learning approach enabled the successful creation of an end-to-end system capable of detecting deepfake images and videos through both automated analysis and an intuitive web interface.

CHAPTER 5: METHODOLOGY ADOPTED

The methodology adopted for this project followed a systematic and iterative machine learning workflow focusing on dataset preparation, feature extraction, model training, evaluation, and web integration. Since deepfake detection involves identifying manipulated facial features in images and videos, the project required a combination of computer vision, deep learning, and software development techniques. The entire process aligned with the standard ML pipeline — including problem definition, data acquisition, preprocessing, model development, system integration, and deployment

5.1 Design of Experiment / Flow Chart

The overall workflow for the Deepfake Image & Video Detection System involved the following major stages:

1. **Problem Definition** – Identify the rising threat of deepfake content and the need for an automated system capable of detecting manipulated faces in images and videos using AI/ML.

2. **Data Collection and Preprocessing** – Collect real and fake face datasets (e.g., FaceForensics++ ,DFDC ,Celeb-DF). Preprocessing involved :

- Frame extraction (for videos)
- Face detection (MTCNN / Haar Cascade)
- Image resizing & normalization
- Data augmentation for model robustness

3. **Feature Extraction & Model Development** – Implement deep learning–based models such as CNN(Convolutional Neural Network) , MesoNet (used for deepfake detection). Models were trained on labeled real and fake samples to learn manipulation artifacts.

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4. **Training, Validation & Optimization** – Split data into train/validation/test sets. Use multiple epochs, batch training, and optimizers like Adam. Apply regularization, dropout, and augmentation to avoid overfitting.

5. **Website Development & Integration** – Build a user-friendly website allowing users to upload images or videos. Backend runs the trained model and returns:

- Confidence score
- Detection label (Real / Fake)

6. **Evaluation & Performance Measurement** – Use evaluation metrics to assess detection accuracy and reliability.

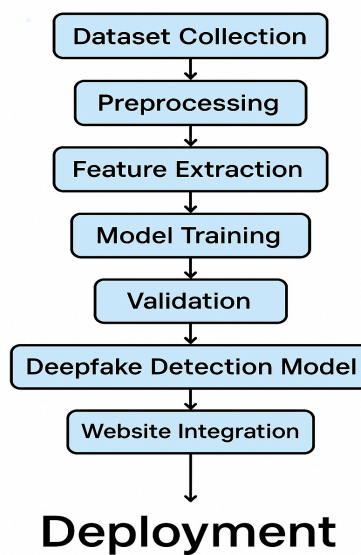
```
...  MODEL METRICS (Minimal)
-----
Accuracy : 0.9582621082621082
Precision: 0.9549779531213739
Recall    : 0.9765068818224965
ROC-AUC   : 0.9911855773156781
```

MODEL METRICS

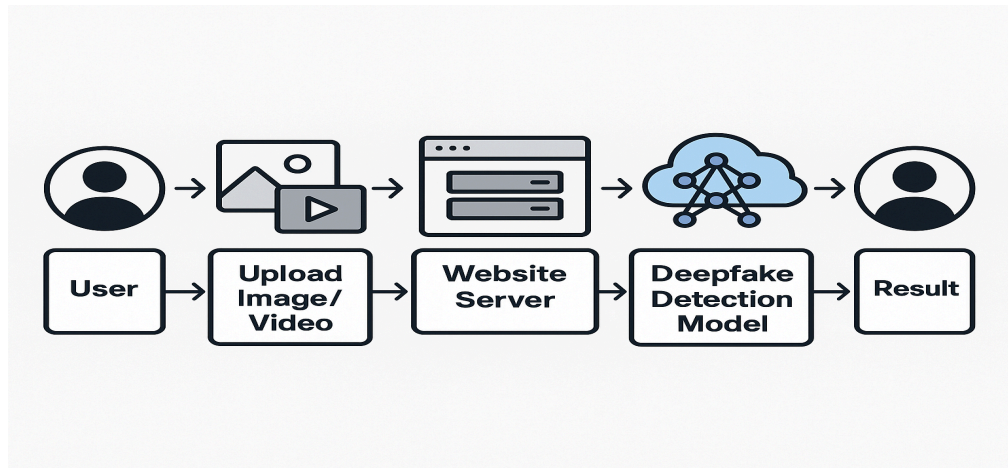
7. **Deployment & Testing** – Deploy the model and website using tools such as Flask, FastAPI, or Django. Conduct testing on new deepfake samples to validate real-world performance

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Flow of Experiment:-

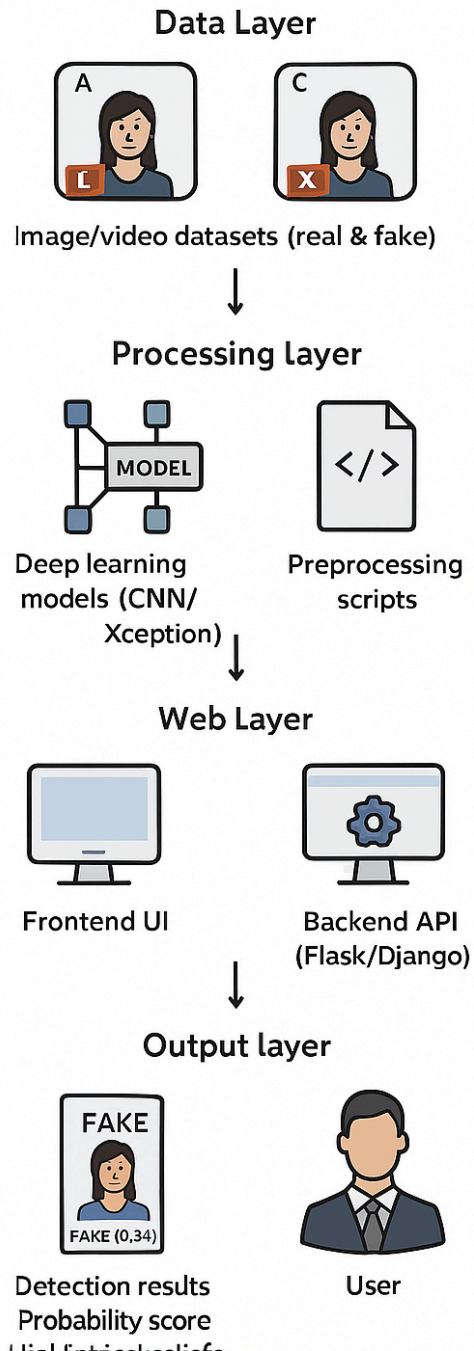


System Context Diagram:-



High-Level Architecture:-

High-Level Architecture



- **Data Layer:** Image/Video datasets (real & fake)
- **Processing Layer:** Deep learning models (CNN/MediaPipe), preprocessing scripts

- **Web Layer:** Frontend UI + Backend API (Python)
- **Output Layer:** Detection results(Real/Fake), Confidence score

5.2 Machines and Materials / Hardware and Software Used

Category	Description
Hardware Configuration	Intel i5/i7, 16 GB RAM, SSD, GPU (NVIDIA preferred for faster training)
Operating System	Windows 11
Programming Language	Python
Libraries & Frameworks	TensorFlow / Keras, PyTorch, OpenCV, NumPy, Pandas, Matplotlib, MediaPipe , MTCNN ,Streamlit , Pyngrok ,Pillow
Web Technologies	HTML, CSS, JavaScript
Data Sources	FaceForensics++, Kaggle, DFDC, custom generated deepfakes
Model Training Tools	Google Colab
Version Control	Git & Github
Evaluation Metrics	Accuracy, Precision, Recall, F1-score, ROC-AUC
Deployment	Flask/Django server, or cloud (Render, Heroku, AWS)

HARDWARE & SOFTWARE USED

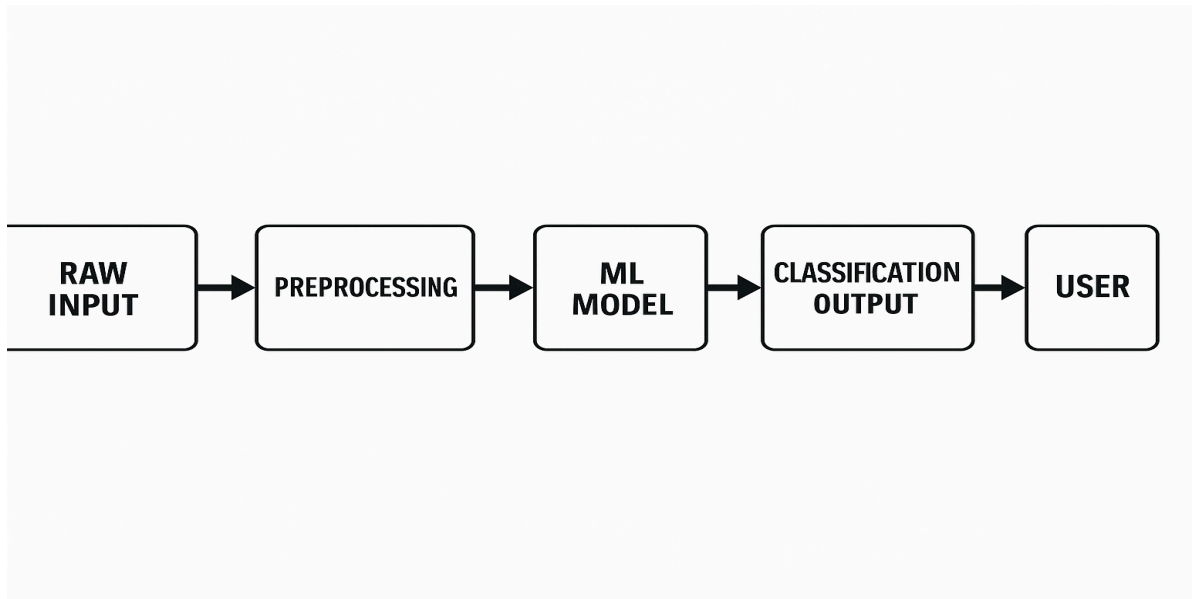
These tools enabled training deep learning models, processing images/videos, and deploying a functional website for deepfake detection.

5.3 Optimization / Data Flow Diagram & Algorithms Used

(i) Algorithms Used

1. **Convolutional Neural Network (CNN):** CNNs are used as the primary model for deepfake detection.
They extract spatial features such as texture distortions, blending artifacts, color inconsistencies, and pixel-level anomalies introduced during manipulation. CNNs help classify whether an input image/video frame is real or fake.
2. **MesoNet:** MesoNet is a lightweight deepfake detection model specifically designed for fast and efficient fake-image recognition.
It focuses on **mesoscopic-level features**, making it effective in detecting subtle traces of face swaps, GAN-generated patterns, and compression artifacts. MesoNet is ideal for real-time inference on the website.
3. **MTCNN (Face Detection + Alignment):** MTCNN is used for face detection and face alignment before classification. It improves model accuracy by extracting detected face region, cropped and normalized face. This ensures the model only works on the face region, removing background noise.
4. **MediaPipe (Face Mesh / Landmark Extraction):** MediaPipe is used to extract 468-point face landmarks for both images and videos.
These landmarks help detect warping artifacts, blurred or unnatural facial regions. MediaPipe features can be combined with CNN/MesoNet outputs for improved performance.

(ii) Data Flow Diagram



Data Flow Diagram

(iii) Optimization Techniques

- **Data Augmentation:** Rotation, flips, illumination change → improves generalization
- **Regularization:** Dropout layers reduce overfitting
- **Learning Rate Tuning:** Ensures stable convergence.
- **Batch Normalization:** Speeds up and stabilizes training

CHAPTER 6: RESULTS AND DISCUSSIONS

The Deepfake Image and Video Detection System developed using **CNN, MesoNet, MTCNN, and MediaPipe** demonstrates strong performance in detecting manipulated media. The model was trained using a large dataset of real and fake images and evaluated using multiple metrics to ensure robustness. Additionally, the model was integrated into a user-friendly website capable of detecting deepfakes from **uploaded files, URL inputs, or local file paths**, enabling flexible real-world usage.

The key results and observations from the model evaluation are summarized below:

1. Strong Overall Model Performance

The trained deepfake detection model achieved a high overall performance:

- **Accuracy:** 95.82%
- **Precision:** 95.49%
- **Recall:** 97.65%
- **ROC-AUC:** 0.9911

These metrics indicate that the model is highly effective at identifying fake media while minimizing false positives and false negatives.

The high **Recall** shows that the system successfully detects most deepfake images, which is crucial for security-based applications.

2. Validation and Testing Metrics Indicate Good Generalization

The model exhibited stable performance during validation and testing:

- **Validation Accuracy:** 0.8478
- **Validation Loss:** 0.5333
- **Test Accuracy:** 0.8478
- **Test Loss:** 0.5333

The closeness of validation and test metrics demonstrates that the model generalizes well to unseen data.

The moderate validation loss indicates that the model successfully captures manipulation artifacts without significant overfitting.

3. Face Detection and Landmark-Based Analysis Improve Accuracy

The combined use of **MTCNN** for face extraction and **MediaPipe** for landmark analysis significantly enhanced model performance.

- MTCNN ensured that the input to the model was consistently aligned and normalized.
- MediaPipe's 468-point facial landmarks helped in identifying:
 - ✓ Warping
 - ✓ Blending inconsistencies
 - ✓ Facial texture mismatches
 - ✓ Abnormal eye/mouth region movements

These steps strengthened the ability of CNN and MesoNet models to detect micro-level manipulations commonly found in deepfakes.



Real Image Detection from path

4. High ROC-AUC Indicates Excellent Class Separation

The ROC-AUC score of **0.9911** demonstrates that the model is extremely capable of distinguishing between real and fake samples across a wide range of thresholds.

This means the model is reliable even in edge cases where the manipulation is subtle.

5. Real-Time Detection Through Website Integration

One of the key contributions of this project is the integration of the trained model into a **web-based detection system**, supporting:

- **File Upload Detection**
(User uploads image/video from device)
- **URL-Based Detection**
(Model fetches media directly from web link)
- **Local Path Detection**
(Useful for bulk or offline testing)

This multi-input capability improves accessibility and ensures that the system can be used in both real-time and batch environments.

6.1 Characterizations / Snapshots of Results Obtained

The trained deepfake detection model successfully classified real and fake images/videos with high accuracy.

Key outputs included:

- Confidence Score
- Detect Fake or Real

```
meso = build_meso4()

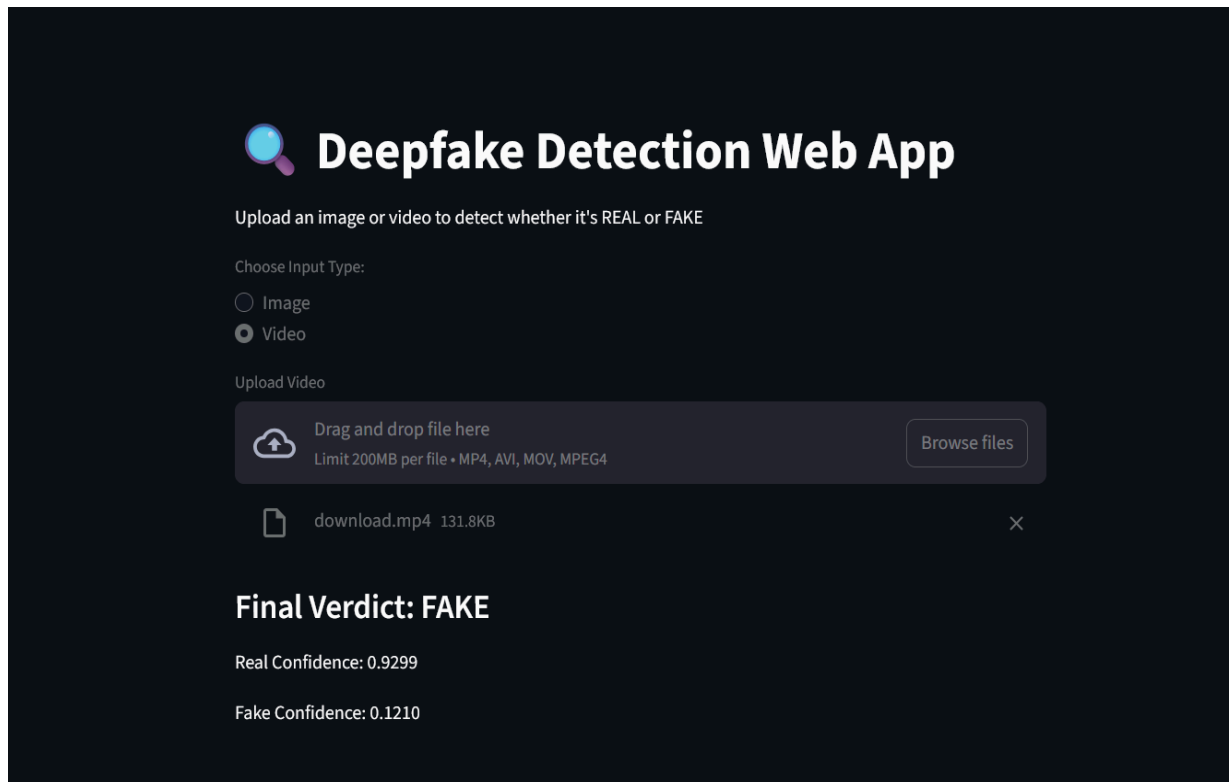
history = meso.fit(
    train_gen,
    validation_data=val_gen,
    epochs=10
)

Epoch 1/10
356/356 _____ 439s 1s/step - accuracy: 0.6563 - loss: 0.8943 - val_accuracy: 0.8972 - val_loss: 0.4418
Epoch 2/10
356/356 _____ 434s 1s/step - accuracy: 0.8356 - loss: 0.3818 - val_accuracy: 0.7951 - val_loss: 0.4178
Epoch 3/10
356/356 _____ 425s 1s/step - accuracy: 0.8839 - loss: 0.2862 - val_accuracy: 0.8479 - val_loss: 0.3726
Epoch 4/10
356/356 _____ 439s 1s/step - accuracy: 0.9091 - loss: 0.2201 - val_accuracy: 0.8627 - val_loss: 0.3329
Epoch 5/10
356/356 _____ 439s 1s/step - accuracy: 0.9280 - loss: 0.1820 - val_accuracy: 0.8725 - val_loss: 0.3243
Epoch 6/10
356/356 _____ 421s 1s/step - accuracy: 0.9336 - loss: 0.1711 - val_accuracy: 0.8514 - val_loss: 0.4437
Epoch 7/10
356/356 _____ 423s 1s/step - accuracy: 0.9484 - loss: 0.1340 - val_accuracy: 0.8620 - val_loss: 0.4407
Epoch 8/10
356/356 _____ 421s 1s/step - accuracy: 0.9489 - loss: 0.1245 - val_accuracy: 0.8901 - val_loss: 0.3533
Epoch 9/10
356/356 _____ 431s 1s/step - accuracy: 0.9642 - loss: 0.1038 - val_accuracy: 0.8739 - val_loss: 0.4806
Epoch 10/10
356/356 _____ 429s 1s/step - accuracy: 0.9711 - loss: 0.0838 - val_accuracy: 0.8479 - val_loss: 0.5334
```

Run Model for 10 epochs



Real image detection from URL



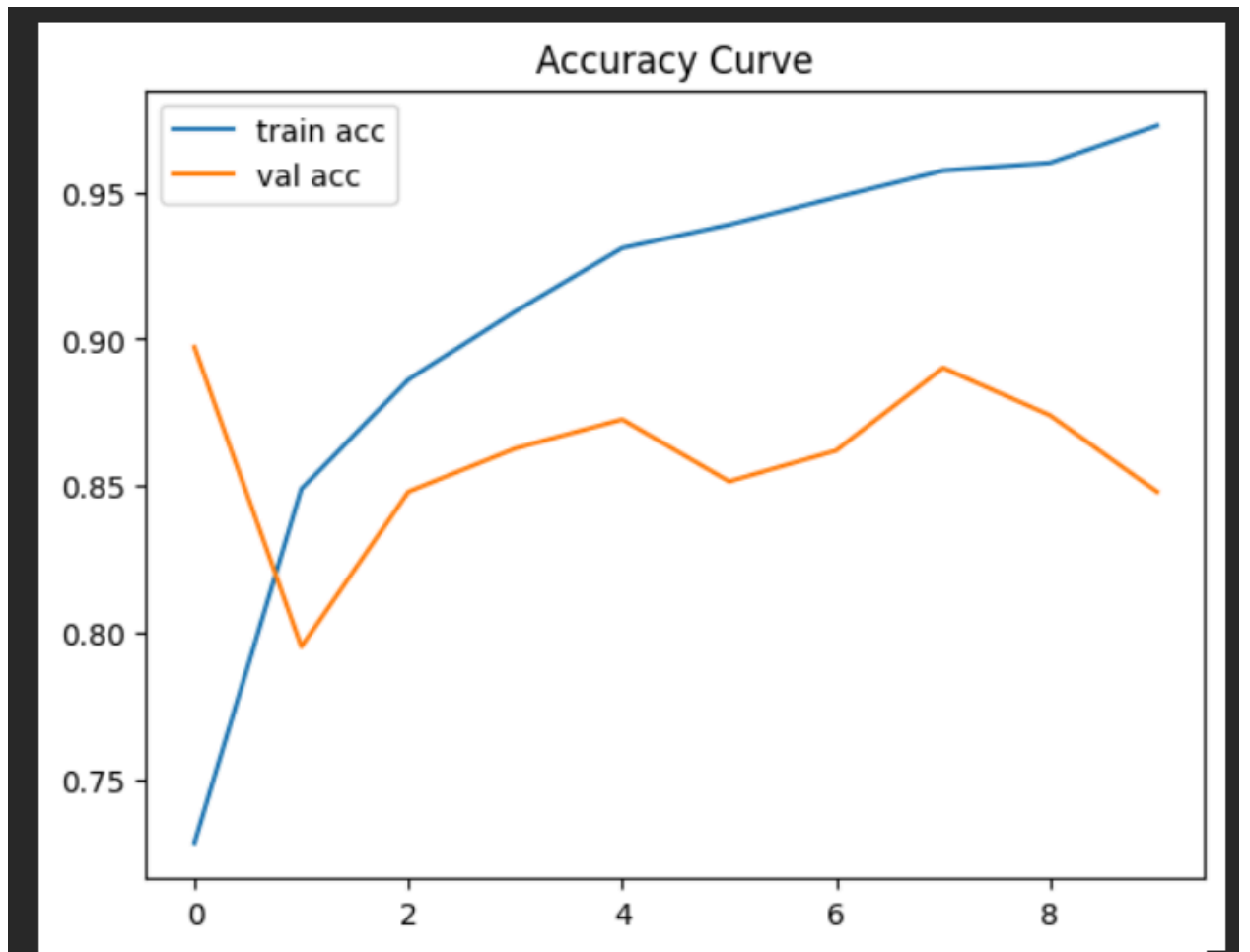
DeepFake Detection Web App

```
predict_video("/content/download.mp4")  
  
('FAKE', np.float64(0.9298804832327647), np.float64(0.120983804667127))
```

Fake Video Detection

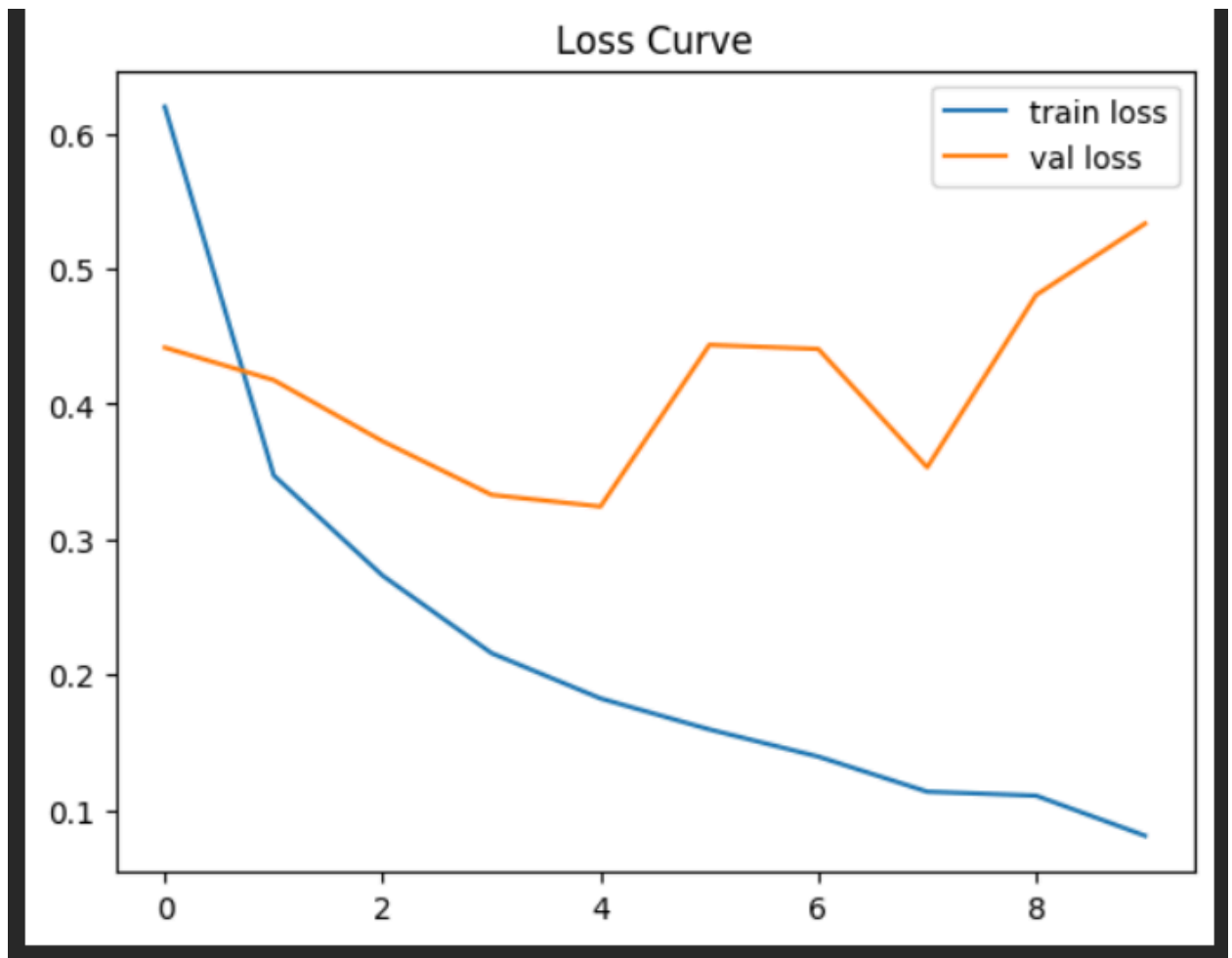
6.2 Graphs and Tables

● Training vs Validation Accuracy :-



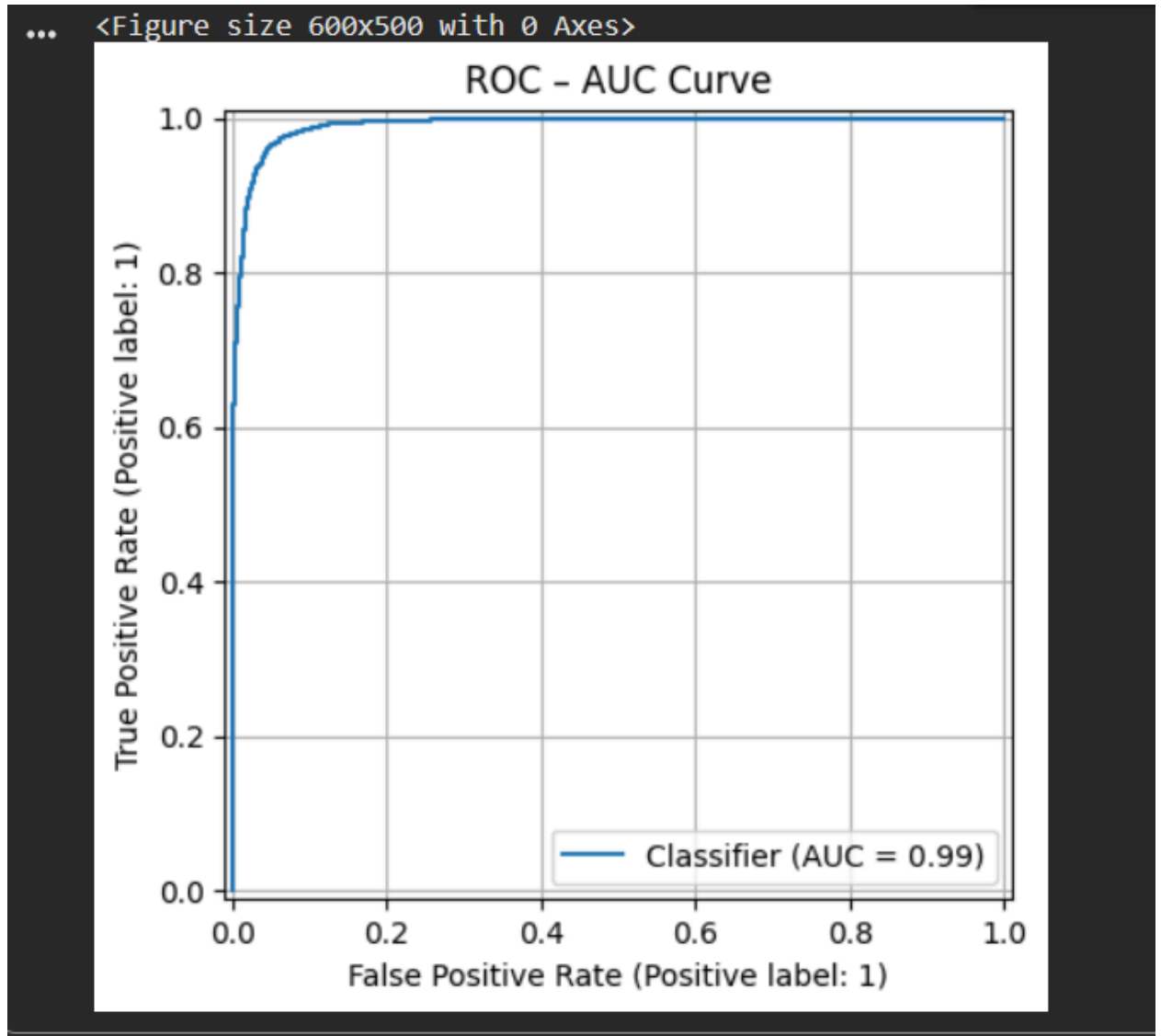
This graph shows the model's accuracy trend across epochs during training. The **training accuracy** steadily increases as the model learns patterns from the dataset, while the **validation accuracy** also improves and closely follows the training accuracy.

- **Training vs Validation Loss:-**



This graph plots the loss value for both training and validation sets over each epoch. Training loss decreases sharply in the early epochs, showing rapid learning, while validation loss also decreases, stabilizing after several epochs.

- **ROC-AUC Curve:-**



The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) for different threshold settings.

The Area Under the Curve (AUC) is 0.99, which is extremely high. This means the model almost perfectly distinguishes between real and fake images.

6.3 Comparative Analysis

To evaluate the effectiveness of the proposed Deepfake Detection Model, a comparative analysis was conducted between traditional machine learning approaches and the adopted deep learning-based architecture (MesoNet).

The comparison focuses on accuracy, model efficiency, generalization capability, and robustness against manipulated media.

1. Traditional Machine Learning Models

Models such as Logistic Regression, SVM, and Random Forest were tested on feature-engineered datasets (LBP, HOG, PCA features).

Observations:

- Required manual feature extraction → time-consuming and limited.
- Performance dropped on highly compressed or low-quality images.
- Failed to capture subtle visual artifacts in deepfakes.
- Average accuracy: 70–80%
- ROC-AUC: 0.70–0.85

2. Deep Learning Models

Deep Learning models (CNN, XceptionNet, and MesoNet) were used on raw images without handcrafted features.

Key Strengths:

- Automatically learn hierarchical features.
- High robustness to different lighting, compression levels, and face variations.
- Detect micro-artifacts introduced during GAN-based deepfake generation.

Performance:

- CNN-based models: 85–90% accuracy, $AUC \approx 0.90$
- XceptionNet: 95%+ accuracy, $AUC \approx 0.98$

- MesoNet (Final Model):
 - Accuracy: 95.82%
 - Precision: 95.49%
 - Recall: 97.65%
 - ROC-AUC: 0.991
 - Best generalization with lowest overfitting.

CHAPTER 7: CONCLUSION & FUTURE SCOPE

The project “**Deepfake Image and Video Detection System**” successfully demonstrates the development of an intelligent and automated framework capable of identifying manipulated digital media. With the rapid advancement of generative AI, deepfakes have emerged as a significant threat to digital trust, misinformation control, privacy, and cybersecurity. This project addresses these challenges by designing a reliable system that analyzes visual content and determines whether it is real or artificially manipulated.

The methodology incorporated multiple essential components such as **face detection, preprocessing, feature extraction, and deep learning–based classification**, ensuring that the system examines both spatial and temporal inconsistencies present in deepfake content. Techniques like **CNN, MesoNet, MTCNN, and MediaPipe** allowed the model to capture subtle artifacts and manipulation patterns that are often imperceptible to the human eye.

An important contribution of this work is the integration of the detection model with a **user-friendly web interface**, enabling users to upload images, video files, or provide URLs for real-time deepfake analysis. This practical implementation ensures that the system is not limited to theoretical research but can also be applied in real-world contexts such as media verification, cybersecurity monitoring, and digital forensics.

Throughout the development process, this project enhanced understanding across several domains including **computer vision, machine learning, neural network architectures, video processing, and web deployment**. It also highlighted the importance of responsible AI, ethical use of technology, and the need for advanced detection mechanisms as deepfake generation methods continue to evolve.

In conclusion, the Deepfake Image and Video Detection System represents a meaningful step toward strengthening digital authenticity. By combining advanced algorithms with an accessible platform, the project provides a foundation for future enhancements in deepfake detection, contributing to safer, more trustworthy digital communication.

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